

CS224: Assignment 6

Weightage: 10%

Date of Demonstration: 1st April 2025 (2PM-5PM) in H/W Lab.

Problem Statement: Implement Max/Min Finding Algorithm in Verilog and demonstrate in FPGA.

Description:

Read data from input and store in a memory. Read one data in one clock. Assume the address is 4 bits and data is 5-bit width. In each clock, mention the address where the data to be written and the actual value to be written. There another signal DataValid indicating we are providing valid input when it is 1. When there is no further input to be read, make it 0. Internally, maintain a count of number of input read. Please note that memory will be accessed internally as well for data processing.

Your objective is to find the maximum and minimum numbers among the inserted data. Implement Maximum and Minimum Algorithms on the data stored in the memory. We should not change the content of the memory at any time. The max/min selection process will start once the reading of the data is completed.

Show the maximum and minimum in the output one by one. There will be outputValid signal to designate that the valid data is shown in the output when it is 1.

Write the memory module and use it. All memory access will be controlled by the write enable and read enable. Consider a single port memory.

Implement a Gate level datapath. For example, if you compare two numbers, implement a gate level comparator module and use it. Behavioral level implementation is not allowed in this Assignment.

Show the Demo in FPGA.

The interface MaxMin(wAddress, DataIn, dataValid, outValid, outAddress, dataOut)

input wAddress[3:0], //address where dataIn to be written

input [4:0] dataIn, //data to be written at wAddress location

input dataValid, //will be on when we want to write input data to the memory

output outValid, //is 1 when sorting is done and valid data is showing at output

output [3:0] outAddress, //address corresponding to dataOut

output [4:0] dataOut //output data corresponding to outAddress location.

In FPGA demonstration, the clock must be taken from outside – manually or from automatic clock. Since you need to show the output of each clock, the automatic clock must be slowed down to KHz level for demonstration. There will be higher marks for automatic clock.

Evaluation:

1. **The Verilog Implementation. The quality of the implementation.**
2. **The Test bench covering various scenarios.**
3. **The FPGA demonstration**

Guidelines:

- All the member of the group need to be present at the time of Demonstration of the assignment. All the absent members will be awarded 0 marks for the assignment. Please show your ID card at the time of demonstration.
- The grading will be based on (a) Correctness, (b) Quality of design, (c) resource*latency optimization, (e) Cleanliness/modular design, (f) Use of proper Comment/Naming/Labeling of the design components and (g) Questionnaire and explanation and (h) correctness of your Verilog implementation.
- The Verilog implementation will also be checked during evaluation.

Submission:

You need to submit following items in a zip folder. For each group, one of the members must submit the report. One submission per group. We will share the marks in teams against your submission each Assignment.

1. A PDF file containing (i) clear diagram of the design (ii) Logic behind the circuit obtained. The report must be available during Lab evaluation.
2. Verilog code of your design
3. Testbench file